

Title

Pitted Terrains on Mars: Modeling High Priority Targets in the Search for Life

Abstract

Recent high-resolution images of the Martian surface collected by the High Resolution Imaging Science Experiment (HiRISE) have captured enigmatic pitted structures within impact craters, interpreted to be the remnants of post-impact hydrothermal systems. These hydrothermal systems are theorized to be transiently habitable systems on the surface of Mars, an otherwise cold and nutrient-depleted planet, indicating pitted structures may be unique high-priority targets in the search for extinct and extant life. However, the dynamics of hydrothermal circulation is uncertain owing to the complexity of fluid flow and heat dissipation within vent pipe structures, making it challenging to define habitable regions on Mars. We propose a novel approach to model steam escape vent pipe formation, using the finite element method (FEM) modeling software COMSOL Multiphysics. The advanced capabilities in modeling heat transfer and fluid flow offered by COMSOL Multiphysics have not been applied to the study of Martian impact crater hydrothermal systems. The proposed models will thus improve our understanding of these structures and validate their status as primary targets in the search for habitable terrains. The results of this proposed program will be critical to ongoing research on the Martian surface, including informing future sampling missions and selecting optimal targets for astrobiological exploration.

- **Scientific Justification**

Modeling Pitted Terrains Will Permit a Habitability Assessment of Mars

The High Resolution Imaging Science Experiment (HiRISE) has captured enigmatic pitted structures on the floors of Martian impact crater Tooting, interpreted to be the remnants of an impact-generated hydrothermal system. Pitted terrains within crater floors, terraces, and impact ejecta are theorized to form through the escape of volatiles, which entrain and deposit material on the surface in pits [1]. Pitted terrains can be used as a proxy for crustal volatile content, such as subsurface water ice [2]. The presence of transient liquid water within impact craters, paired with residual high temperatures due to impact processing, can form impact-generated hydrothermal systems (IGHSs). These systems have been proposed as promising targets in the search for life [3]. Therefore, IGHSs are unique high-priority targets in the search for extinct and extant life on Mars, an otherwise cold and nutrient-depleted planet.

Defining habitable regions within impact structures on Mars is challenging owing to the unknown dynamics of fluid flow and heat dissipation within steam escape vent pipe structures [1-4]. This problem can be addressed now, given access to detailed topographic analyses permitted by HiRISE imaging, and the highly sophisticated heat transport and fluid flow modeling capabilities afforded by COMSOL Multiphysics. It is critical to address this question due to priorities outlined in NASA's *Origins, Worlds, and Life* Decadal Survey, which highlights assessing Martian habitability [5]. This modeling program will address these themes and offer insights into the selection of astrobiology targets for future study.

We propose to model the process of steam escape through Tooting Crater using COMSOL Multiphysics, to investigate the formation of pitted terrains. This program is directly relevant to the NASA ROSES solicitation, and shares priorities with NASA's science mission directorate (SMD). The ROSES program calls for the "(t)heory, modeling, and analysis of SMD science data..." which this proposal addresses, as well as calling for research to "... explore the Solar System to study its origins and evolution, including the origins of life within it (Appendix C)." This program will offer a deeper understanding of volatile content within the Martian crust in Amazonis Planitia, and insights into the habitability of impact generated hydrothermal systems. Results will include a novel COMSOL modeling framework which can be applied to other IGHSs on Mars, expanding the scope and impact of this modeling program.

Unconstrained Steam Escape Dynamics Inhibit Habitability Assessments

Pitted structures within Martian impact craters form through the ejection of subsurface material during volatile degassing through impact melt. This material is entrained within rapidly rising steam and deposited on the surface to form a network of debris mounds and pits, which are interpreted as steam escape vent pipes [1]. Therefore, the presence of pitted terrains within impact craters indicates the presence of water within crustal material prior to impact processing [2]. Tooting Crater contains morphological evidence of water-rock interaction in addition to pitted terrains, such as outflow channels [4], corroborating the presence of subsurface volatiles. The presence of volatiles in

impact craters can produce IGHSs, which have been posed as transiently habitable regions on the surface of Mars, with access to heat and liquid water [3]. However, unconstrained volatile dynamics within steam escape structures makes assessing the habitability of IGHSs challenging. Untangling these dynamics with estimates of volatile escape speed, duration of degassing, and pre-impact crustal volatile content will allow for estimates of the habitable timescales of IGHSs, providing limitations on the potential for microbial colonization. Modeling with COMSOL Multiphysics finite element method (FEM) software will provide much-needed constraints on the dynamics of pitted terrain formation, allowing for quantitative assessments of impact-generated hydrothermal system habitability within Tooting Crater.

This Proposal: Modeling Vent Pipe Structures Will Permit a Habitability Assessment

We propose to model the dynamics of steam escape through impact melt deposits within Tooting Crater, using the FEM software COMSOL Multiphysics. Assessing the habitability of impact-generated hydrothermal deposits is well suited to NASA ROSES program priorities, as “(t)he Planetary Science Research Program sponsors research to explore the Solar System to study its origins and evolution, including the origins of life within it (Appendix C) [6].” In addition, NASA’s *Origins Worlds and Life* decadal survey prioritizes investigating “dynamic habitability,” calling for research addressing “(w)here in the solar system do potentially habitable environments exist, what processes led to their formation, and how do planetary environments and habitable conditions co-evolve over time,” questions tackled by this modeling program [5].

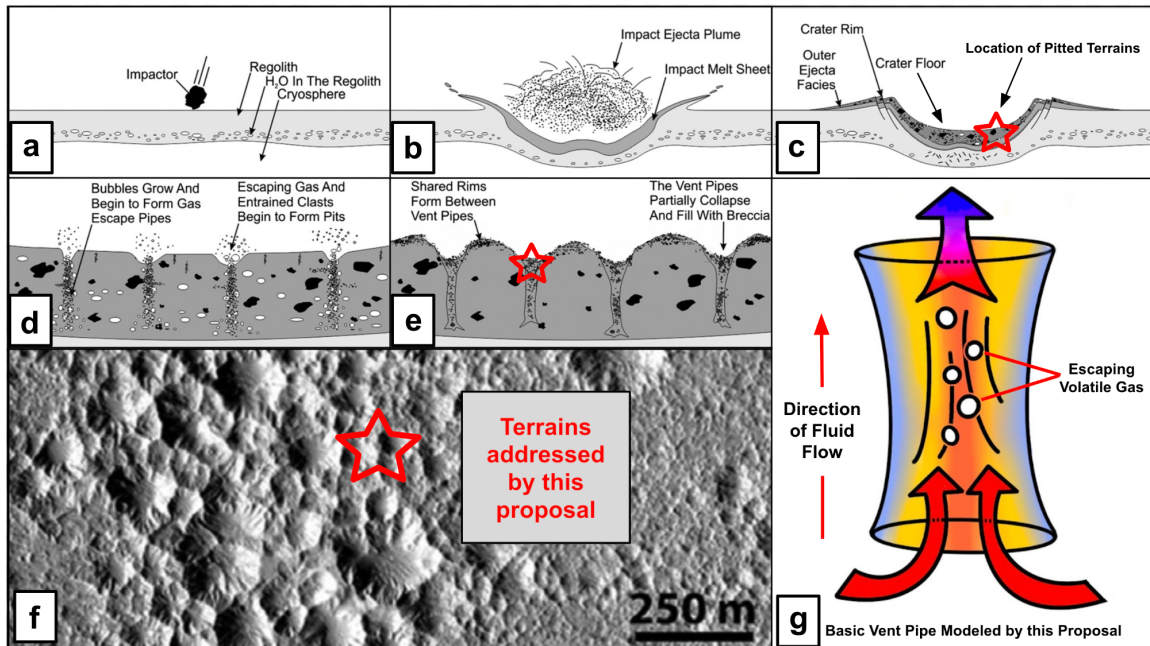


Fig 1. Constraining the dynamics of heat flow within Martian IGHSs is crucial to assessing the microbial habitability of these terrains. Panels a through e, modified from Boyce et al., 2012, describe the formation of vent pipes within impact melt deposits. Panel f is from HiRISE image PSP_002158_2035 depicting pitted terrains on the floor of Tooting Crater, with red stars

denoting areas with pitted terrains. Panel g is a schematic illustration of a vent pipe structure to be modeled by this proposal, with arrows indicating the direction of flow through the conduit. Our understanding of the habitability of impact crater hydrothermal systems is incomplete, given the formation of pitted terrains has not been studied with modern FEM approaches. We propose to model volatile escape through vent pipe structures using the FEM software COMSOL Multiphysics, which will constrain heat and fluid flow within the Martian impact crater Tooting. With these models, we will ascertain the duration of post-impact habitable temperatures and water volumes, informing future mission sampling strategies which prioritize astrobiologically relevant terrains.

This program will reconstruct a symmetrical vent pipe structure (conduit) within 3D space in COMSOL Multiphysics, illustrated by Fig. 1 panel g, and utilize the (1) heat transfer in porous media, (2) heat transfer in solids and fluids, and (3) conjugate heat transfer modules to analyze steam escape within the system. This program will provide constraints on the speed of steam escape required to produce observed surface features through particle entrainment and transport, the pre-impact crustal volatile content of the Tooting Crater impact site, and the duration of elevated temperatures in the post-impact hydrothermal system. The model will be validated against topography derived from HiRISE imaging, to ensure resulting features match observed pitted terrains within Tooting Crater. These quantitative constraints (escape speed, crustal volatile content, and system duration) will permit a novel evaluation of the habitability of impact-generated hydrothermalism within Tooting Crater.

Broader Impacts: Assessing Habitability as a NASA Priority

Constraining the habitability of IGHSs will permit a deeper astrobiological investigation of these structures, along with improved sampling site selection strategy for future astrobiology missions to Mars. This modeling program fits into the *Origins, Worlds, and Life* Planetary Science Decadal survey priorities, investigating habitability questions central to the agency's themes and goals [5]. Modeling the habitability of Martian IGHSs directly addresses the NASA ROSES funding call, and agency science mission directorate (SMD) through the assessment of Martian habitability [6]. This critical research is invaluable to the community, and modeling products will advance research in Martian habitability. The modeling conducted in this program for Tooting Crater will serve as a blueprint for the assessment of impact-generated hydrothermal system habitability in multiple craters across the Martian surface, vastly expanding the scope of the Martian surface for which it is possible to constrain habitability. Additional modeling products will include new insights into the volatile content of the Martian crust, with implications for climate modeling, as well as a deeper understanding of physical impact processing on Mars.

References

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